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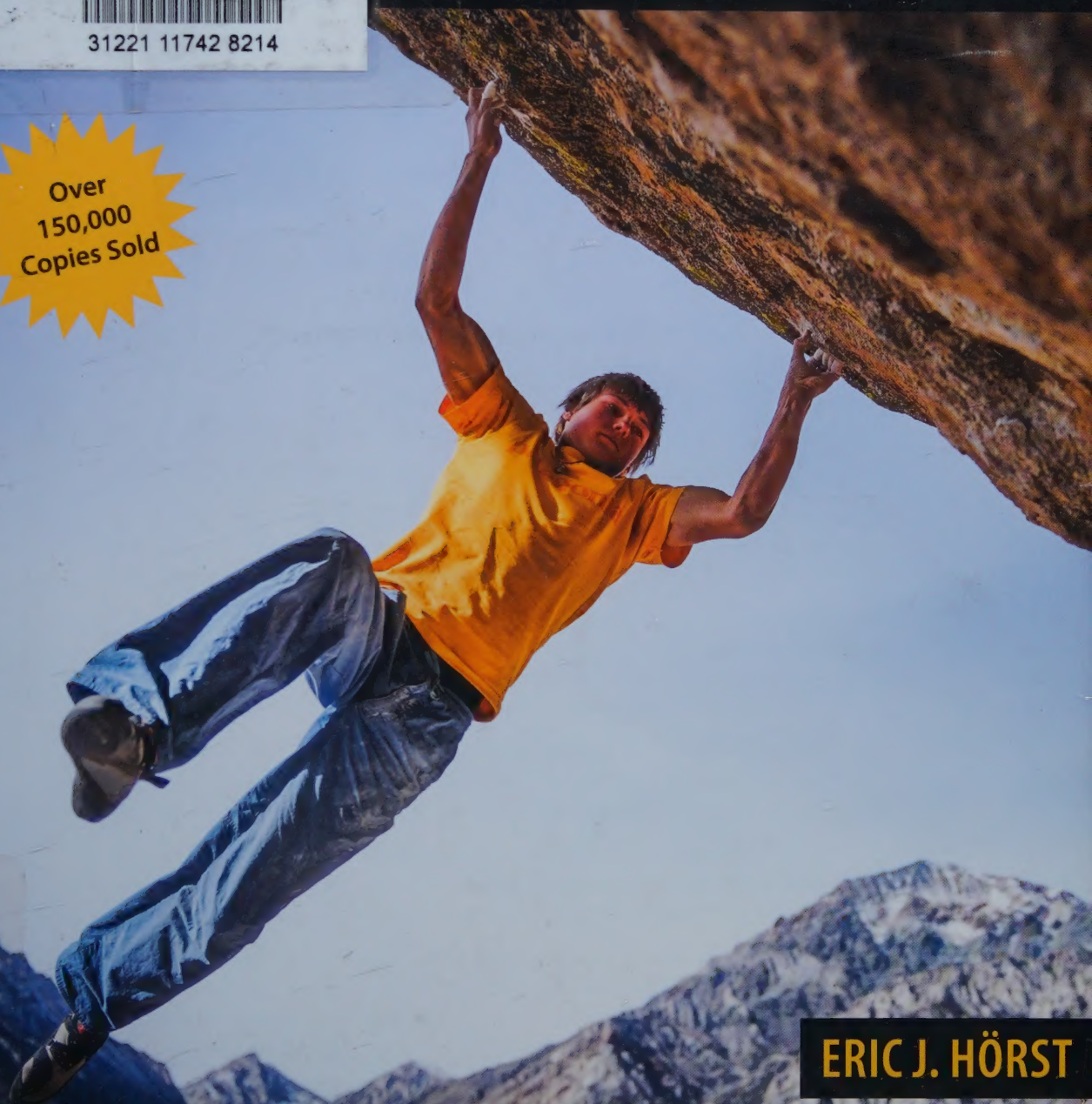
The Definitive Guide to Improving Your Performance

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ERIC J. HÖRST

Praise for the second edition of *Training for Climbing*

“Covering everything from mental exercises to injury prevention to nutrition to goal-specific workouts, *Training for Climbing* is a must-read for climbers of any ability who are looking to take their performance to the next level.” —*Climbing Magazine*

“Eric Hörst has taken complex scientific training concepts and integrated these with contemporary research on climbing to produce a very readable and useful training guide. This book will lead the climber to new heights of performance.” —Phillip B. Watts, PhD, exercise physiologist and climbing researcher, Northern Michigan University

“Hörst is uniquely positioned to bring current methods in sports psychology and exercise science to the world of climbing, and he has hit the mark in superb style! If you are passionate about climbing and getting better, *Training for Climbing* will become your dog-eared companion!” —Richard K. Fleming, PhD, assistant professor of psychology, University of Massachusetts Medical School

Reader Reviews from Around the World

“I would like to thank you for writing *Training for Climbing*! It has been incredibly helpful and inspiring, and has helped me improve quickly. I’ve recommended it to everyone on my climbing team!” —Sarah (New Hampshire)

“Thanks for writing *Training for Climbing*; it’s an amazing book!” —Alex (Spain)

“Hello Eric, thank you for all the time and energy spent researching and writing. I purchased *Training for Climbing* when it first came out and within a year I was climbing 2 grades harder. We call it the ‘cible.’” —Cody (Utah)

“Having never climbed, at age 26 I dove into climbing and thanks to *Training for Climbing* I lost 15 pounds and got myself to 5.10s in just four months. I sincerely appreciate the specific attention you give to mental training and what I like to call ‘enlightened climbing.’ Your writing is beautifully clear.” —Paul (Michigan)

“*Training for Climbing* is a great book—it is testament to a sport filled with nuance. I love learning as much as I can about every aspect of this sport, and your books make me want to get out there and climb!” —Chris (UK)

“Let me say that I am a HUGE fan of your writing. Your books *Training for Climbing*, *Conditioning for Climbers*, and *How to Climb 5.12* have changed my life and approach to training for the better!” —John (Massachusetts)

“Hey Eric, just to say thanks for your book *Training for Climbing*. I have read it more than once and every time I find some new valuable info for improving my climbing.” —Felipe (Guatemala)

"I'm a huge fan of your climbing training products, books, and training programs. If only I had found them sooner!!" —Jake (Virginia)

"I have been reading your recent book *Training for Climbing* (twice, cover to cover) and I'm now on my way to rescheduling my training with a much more nuanced focus than before. Thank you for the enlightenment and inspiration!" —Sigurd (California)

"I want to tell you that *Training for Climbing* is clearly the best and most comprehensive book on climbing training methodology I have ever read. In particular, the chapters on mental training and skills/strategy training have inspired me and given me extremely valuable insights for my future workouts. Thanks for such a great work!" —Guido (Switzerland)

"I have just read your book *Training for Climbing*, and I think it is great!" —Sven (Croatia)

"Eric, just wanted to say thanks for writing *Training for Climbing*. Since I started reading it and training, I have improved tremendously. I am having a great time doing the strength-training exercises and just enjoying the process of climbing with a different mentality." —Carlos (California)

"Thank you, Mr. Hörst, for helping me achieve my first 5.12a just a few days ago!" —Hiroyasu (Japan)

"Hi Eric, I've just finished reading *Training for Climbing*—great work! Over the last fourteen years I have studied climbing, and to date your book is by far the most accessible to average climbers." —Paul (Australia)

"I read a quote from your training book and consequently bought it. This book has changed my whole attitude in climbing and life due to your mental training chapter. Just wanted to say thank you!" —Scott (Pennsylvania)

"Thanks for your great books, Eric. I have read them all—they are the best!" —Ofer (Israel)

"What a great book! Your HIT workout definitely gave me much more strength. Thank you very much for your great work, Eric!" —Andre (California)

"Hi Eric, I read your book *Training for Climbing*. I think it is the best book in the field." —Roberto (Ecuador)

HOW TO CLIMB™ SERIES

Training for Climbing

The Definitive Guide to Improving Your Performance

Third Edition

Eric J. Hörst

Foreword by Phil Watts

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This is an instruction book about rock climbing, a sport that is inherently dangerous. Do not depend solely on information from this book for your personal safety. Your climbing safety depends on your own judgment based on competent instruction, experience, and a realistic assessment of your climbing ability.

The training and nutritional advice given in this book are the author's opinions. Consult your physician before engaging in any part of the training and nutritional program described by the author.

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Bohica (5.13b), Red River Gorge, Kentucky

*To two of the strongest, most innovative
and humble men ever to pull down on
rock: the "Master of Rock" John Gill
and the late, great Wolfgang Güllich.*



***Nina Williams
crimping up
Baboon Platoon
(V8), Rocklands,
South Africa.***

BRUCE WILSON

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Foreword

Elite athletes and coaches are well aware of the importance of sport-specific physiological, technical, and mental training for optimizing progress toward high-level performance. Even those of us who are more than a bit distant from elite-level performance can still realize improvement and an enhanced experience from the application of known training principles. Sifting the truth from the full array of information available these days can be difficult. Fortunately, there are a few writers who present contemporary evidence-based concepts in a manner that is easily understood and applied. Eric Hörst is one of these writers.

I have been in communication with Eric since the first edition of *Training for Climbing*. We have discussed the science of climbing via phone, e-mail, and over Miguel's pizza at the Red River Gorge. I am impressed by Eric's consumption and digestion of both rock climbing-specific research and sports science research in general. He has a keen ability to bring the science into application. As a bonus, Eric has insights only a practitioner can have through his more than three decades as a veteran high-end climber and coach.

Effective and optimal physical training is more than the old adages of "break a sweat a couple of days per week" or "no pain, no gain." As an exercise science researcher and educator, I stress the importance of understanding the physiological performance model for a specific activity. With rock climbing, this model includes basic components such as muscular strength and endurance, rate of force development, aerobic and anaerobic power, anatomical range of motion, and energy economy. Specific workout designs for developing each component differ. A climber is challenged by the process of deciding which workout designs to use, how much to do, and how to best recover. When movement techniques and mental aspects, along with proper nutrition, are

added into the mix, the task can be overwhelming. In *Training for Climbing*, Eric has provided workable details in an organized and logical manner and in a way that enables the climber to customize training to his or her specific needs.

I have been in the academic and research areas of exercise science for over forty years and have seen concepts develop and evolve, come and go, and come again. I began in exercise physiology at a time when lactic acid was considered a problematic end product of anaerobic metabolism . . . a "bad" thing to avoid at all costs. Now we appreciate lactate as a dynamic component of high power production that may also serve as an energy source away from the specific site of its production . . . a "good" thing. I also grew up in a time when protein intake was only a concern for "strength" athletes such as football players and weight lifters. Now we know protein intake per body mass unit can be higher in "endurance" athletes during periods of high-volume exercise. The exercise physiology text I used during my master's-level graduate studies was 200 pages and around half a kilogram in weight. The text I currently use with my undergraduate students is over 800 pages and weighs three kilograms! The sheer volume of available information presents a challenge. This new edition of *Training for Climbing* eases the information overload and presents a solid foundation upon which to explore.

Whether you aspire to stand upon the podium at competitions, bump your peak level a grade or two, or squeeze more moves out of limited vacation time on the rock, there are words for you here. By all means read *Training for Climbing*, work out a plan that is a good fit for you and your goals, then get on the rock for the joy of it.

—Phil Watts, PhD, FACSM
Professor of Exercise Physiology
Northern Michigan University

Acknowledgments

Writing a book is an Everest-like undertaking. It is a team effort that can take years to plan and execute, but getting to the summit still takes an immense individual effort and an indomitably singular focus.

Having completed this climb (more than a few times!), I reflect on all that has brought me to this point—the thousands of wonderful days I’ve spent on the rock, the countless climbers from around the world I’ve had the pleasure to meet, and the dozens of partners I am grateful to have shared a rope with—and my recent climbing adventures with my badass sons. Learning to climb is a long, continuous process with no end, and I thank all the people who have influenced me from my days as a wide-eyed teenage rock jock to a fifty-something veteran climber. Specifically, I must thank directly a few key players who inspired me as a climber and coach: John Gill, the late Wolfgang Güllich, Lynn Hill, John Long, Jim McCarthy, Pat Ament, Richard Goldstone, Mark Robinson, the late and legendary Todd Skinner, and Tony Yaniro—all of you, knowingly or unknowingly, have inspired me and contributed to this book in some way.

Though I continue to train and climb with a passion similar to my younger days as a national-class climber, I enjoy more every year the role of climbing coach and researcher. I am humbled by all the letters and e-mails I’ve received from climbers in more than fifty countries who have read my books, magazine articles, and Internet posts. I appreciate all the feedback, the suggestions, and, most of all, the success stories of those who have benefited from my work. Writing climbing books is certainly not a lucrative endeavor, but knowing that I’ve helped thousands of people from around the world climb better is priceless.

Creating this third edition of *Training for Climbing* turned out to be another massive undertaking.

It’s been an eight-year journey since the last edition, and during this time I’ve read hundreds of research papers and integrated the key findings into my training programs and this book. And lucky for me (and you), I have a dream team of great climbing minds to discuss new concepts and training applications—many thanks to my PhD’d and coaching friends Simon Fryer, Phil Watts, Eva Lopez, Rick Fleming, Richard Schmidt, Jim Sullivan, Natasha Barnes, Steve Bechtel, Dicki Korb, and Patrick Matros. Thanks for sharing your unique insights!

I’d like to extend a sincere thank you to John Burbidge, Julie Marsh, and everyone at Falcon-Guides and Globe Pequot who helped bring this book to fruition. I am also very appreciative of all the climbing companies that support me and my many projects, including Nicros, La Sportiva, Maxim Ropes, DMM/Excalibur, Organic Climbing, and Köhl. I am most grateful for the input and participation of Kris Hampton, Hannes Huch, Anna Davey, Chadd Gray, Jessica and Geoff Britten, Alex Megos, Crystal Norman, Jessica Rohm, and the gyms Earth Treks, Vertical Endeavors, Philly Rock Gym, and Spooky Nook. Mega kudos to Claudia Zeigler, Savannah Cummins, Matty Hong, Chris Noble, Kyle Duba, Thomas Ballenberger, Phil Bard, Gerd Heidorn, Andrew Chao, Theron Kirkman, and Bruce Wilson, whose photography is featured throughout the book. Heartfelt thank yous to my parents, Bob and Ethel, my sons, Cameron and Jonathan, and my wife, Lisa, for joining me in my training and climbing and for teaching me so much—I love you each more than you can imagine.

Finally, I am sincerely thankful for my original climbing partner and real-life hero, Jeff Batzer. Despite losing five fingers and half a leg, you still possess—and show by example—the power of climbing. Jeff, you are a true inspiration—thanks, man.

Introduction

*A man's reach should exceed his grasp or what's
a heaven for?*

—Robert Browning

Training for Climbing is a unique synthesis of more than three decades of studying, researching, imagining, and experimenting with new ways to increase climbing performance. Building on the foundation laid out in the first two editions of *Training for Climbing*, this massively revised and expanded third edition of *TFC* establishes a new benchmark for cutting-edge, comprehensive, and evidence-based expert instruction on the subject of elevating climbing performance.

Training for Climbing presents a success strategy based on leading-edge sports science and the latest climbing research that's amplified by powerful practice and training techniques used by Olympic athletes (but unknown to many climbers). If you commit to dovetailing your physical training with regular use of the potent mental-training techniques described herein, you will become a far better climber regardless of your present ability. By faithfully applying just 50 percent of the methods contained in this text, you will surely grow to outperform the mass of climbers. And if you integrate most of the material into your training, climbing, and daily life, you may very well progress to a level beyond your current comprehension!

Training for Climbing is as much about developing new ways of thinking as it is about engaging in new ways of training. A common thread that weaves throughout this book is that you must train smarter to climb harder; intelligence in climbing is not measured by IQ, but instead by the quality of your thoughts and actions. The thoughts you carry and the things you do (or don't do) are ultimately what separate you from the mass of climbers. Whether

you flash or fall, become superstrong or get injured, or feel happy or frustrated, springs forth from subtle differences in the ways you think, feel, and act compared with other climbers. Therefore, the primary goal of this book is to help guide you to more deliberate and effective ways of thinking and acting—both training and non-training actions—in your pursuit of peak climbing performance.

Since climbing is all about an intimate dance between you and the rock, it's vital to recognize that your climbing performance evolves from the inside out, and that you only slip and fall when you blow a move. Goethe wrote, "Nature understands no jesting; she is always true, always serious, always severe; she is always right, and the errors and faults are always those of man. The man incapable of appreciating her, she despises; and only to the apt, the pure, and the true, does she resign herself and reveal her secrets." From this perspective it becomes obvious that we must always look inside ourselves to determine the cause of our struggles. Looking outward for the reason or to place blame is a loser's game.

This book begins with a logical progression of self-analysis, goal setting, mental training, and technique training before you ever lift a weight. Chapters 1 through 4 are focused on helping you learn, most quickly, the vital mental and technical skills that separate the best of climbers from the rest of climbers. After a brief review of the history of training for climbing, you'll get started on the road to better climbing by taking a self-evaluation test that will reveal your true strengths and weaknesses on the rock. Armed with this diagnosis, you can effectively address your limiting constraints, whatever they might be. If they are technical or mental in nature, then there's lots of gold to be mined from the pages of chapters 3 and 4.

Chapters 5 through 10 present the most in-depth look at strength training for climbing ever published. As an intensely practical person with a background in math and science, I have always felt it important to delve into the theory and application of cutting-edge sports science—after all, knowledge is power. Accordingly, chapter 5 provides a 200-level understanding of the exercise physiology of training for climbing. Read slowly and carefully, and you'll discover many training secrets and nuances that, if applied steadfastly, will make your training more effective than likely anyone else you know. Chapters 6 and 7 are brand new to this edition of *TFC*, and they present essential stabilizer, core, and posterior chain training that will make a huge difference in your climbing and help reduce injury risk as you move into the higher grades. Chapters 8 and 9 provide dozens of climbing-specific exercises for increased strength, power, and endurance in the important finger flexor and pulling muscles. Then, in chapter 10 you'll learn how to craft a high-quality training program and integrate it with the rest of your everyday activities.

The concluding chapters 11 through 13 cover the often overlooked subjects of performance nutrition, recovery, and proper treatment and prevention of common climbing injuries. Becoming a complete climber requires that you embrace these subjects with the same fervor as you execute your training program or plot your next climb.

Performing in the vertical extreme requires strength, power, and endurance; vast technical skills; and the ability to focus and manage fear in potentially dangerous situations. In aggregate, these factors make climbing one of the most complex sporting activities on this third rock from the sun.

Throughout the text I have cited select research in case you'd like to peruse the nitty-gritty details—these references and other useful information can be found in the back of the book.

To glean the greatest benefit from *Training for Climbing*, I encourage you to employ active reading techniques such as highlighting key passages, putting a star next to the most meaningful strategies, and taking notes for later review. Consider lifting some of the most powerful phrases and posting them in places where you'll see them throughout the day. Review your highlighted passages and notes every few weeks, and reread the entire book in six months to a year. Not only will this reinforce your understanding and mastery of the concepts, but you also will gain new insight and distinctions as you become a different person at each read-through.

As you cast off on your training-for-climbing journey, I want to wish you much strength and power, as well as many grand, transcending days as you climb through this world of wonder. Though we may never meet, we are connected through our shared passion for climbing. I welcome your feedback, and I look forward to hearing from you after some grand success that undoubtedly awaits you. Here's wishing you many safe and wonderful days on the rock!

—Eric J. Hörst
www.Training4Climbing.com



*Adam Ondra on the first
ascent of Move (5.15b/c),
Flatanger, Norway.*
CLAUDIA ZEIGLER